Formula 1 Mission Control

Racing on Probability, Not on Plan

A prescriptive analytics teaching case | Simulation module | Core MBA

Hossein Piri and Saed Alizamir

Fifty laps, one decision

The numbers on Maya Cardoso's screen had stopped being numbers an hour ago. They were now a feeling in her chest. It was the end of lap 23. Two laps from the only pit stop her car would make all afternoon, she had a choice to make, and roughly eight seconds in which to make it. Around her, the room they call Mission Control was almost silent. The decisive analysis was not happening beside the car. It was happening here, in a windowless room at the factory, while the car threw itself around a circuit three thousand kilometres away at three hundred kilometres an hour. The same data the car generated arrived on Maya's screen delayed by about a tenth of a second.

The plan was simple and the plan was good. Pit on lap 25, the exact middle of the race. Fresh tires for the second half, stints balanced, the math clean. Her engineers had run the optimization a hundred times and it always returned the same answer. Lap 25. Pit on plan.

But Maya was not looking at the optimization. She was looking at a probability that had been creeping upward for ten laps: the chance that a safety car was about to come out. And if it did, everything she thought she knew about lap 25 would be wrong.

The business of going around in circles

It is tempting to think of Formula One as a sport. It is more accurate to think of it as a business that happens to be conducted at speed. Strip away the noise and what remains is an environment that any executive would recognize: low latency, data rich, heavily regulated, cost capped, publicly visible, fiercely competitive, and unforgiving of mistakes. The particular problem this business solves is making a car travel a fixed distance faster than ten rival cars. The methods it uses to solve that problem travel surprisingly well into boardrooms.

Three things shape the work. First, a cost cap. Formula One introduced a cost-cap regime in 2021, so success now depends less on out-spending rivals and more on converting a fixed budget into performance. Second, strict limits on track testing. A team cannot simply take its car out whenever it likes to try things, so it must learn much of what it needs from simulation. Third, and most important for Maya this afternoon, a hard truth about uncertainty. Some things the

team can know with great precision, such as how its own car will behave. Other things it cannot know at all, such as when a rival will crash and bring out a safety car. The art of race strategy lives entirely in that second category.

Why the team simulates instead of optimizes

Maya's department draws a sharp line between two activities that outsiders tend to confuse. Optimization, as her team uses the word, is deterministic. It assumes the world will unfold in one known way and it returns a single best answer for that world. Lap 25 is an optimization answer. It is correct, and it is correct only in a race where nothing surprising happens.

Simulation is different. Rather than ask what is the best move if the future is known, it asks what range of futures could occur, how likely each one is, and how a given policy would perform across all of them. The team runs these simulations not by the dozen but by the hundreds of thousands per second, watching not for a single answer but for which situations are rising in probability. When a safety car becomes likely, a strategy that ignored that possibility stops being optimal, even though the optimizer never said so.

The reason a safety car matters so much comes down to one mechanism. A pit stop costs time, because the car must slow, enter the pit lane, stop, and rejoin behind the cars it was racing. Under normal green-flag racing that costs the team around twenty three seconds. But when a safety car is deployed, the entire field slows and bunches together. A car that pits during that window loses far less, because its rivals are crawling too. The same physical pit stop can cost twenty three seconds on one lap and a quarter of that on the next. A safety car turns an expensive necessity into a bargain, for whoever is positioned to take it.

Two philosophies on the pit wall

This is the choice in front of Maya, and it is really a choice between two policies.

STATIC, or pit on plan.

Pit on lap 25, under green-flag racing. The stints are perfectly balanced, the tire wear is symmetric, and the outcome barely varies from race to race. It is the answer the optimizer loves. It never wins by a lot and it never loses by a lot.

REACTIVE, or stay out and gamble.

At the end of lap 23, do not pit on the planned lap 25. Stay out. If a safety car is deployed before the car reaches its forced stop on lap 30, pit immediately and take the cheaper stop, gaining time on every rival who already pitted at full price. If no safety car appears by lap 30, pit under green-flag conditions on lap 30, now on worn tires and with unbalanced stints, finishing slightly behind what the plan would have delivered. The gamble is simple: lose a little if the safety car does not come, gain a lot if it does.

Maya has seen both philosophies win and both lose. She has also seen something stranger. She has watched a strategist make what everyone later agreed was the right call, and lose the race anyway, because the safety car simply never came. And she has watched a lazy call rewarded by luck. Her team has a discipline for exactly this confusion.

Exhibit 1. Stylized model inputs

Quantity	Value
Race length	50 laps
Decision point	end of lap 23
Planned pit lap (STATIC)	25
Forced REACTIVE stop if no safety car	30
Usable safety-car window	laps 26 to 30
Probability of a usable safety car	35%
Green-flag pit loss	23 seconds
Safety-car pit loss	6 seconds
Fresh-tire lap time	90 seconds
Tire degradation	0.06 seconds per lap of tire age
Execution noise	1.0 second standard deviation

How the team judges a decision

After every race, win or lose, the team holds two reviews. The first is immediate and emotional, a place to put feelings on the table before they distort anything. The second comes days later and is the one that matters. It is calm, data driven, and free of blame. It follows one rule that most organizations never adopt.

The team refuses to assume any link between the outcome of a race and the quality of the decision that shaped it. A good result might have followed a poor decision rescued by luck. A poor result might have followed an excellent decision that the dice simply did not reward. So the team separates three things and evaluates each on its own: the quality of the decision, the quality of the execution, and the outcome. A decision is judged by whether it was the best bet given what could have been known at the time, not by how the single race happened to turn out.

Your assignment

Maya's team evaluates race strategy by expected total race time across many comparable futures. Lower time is better. But Maya also knows that her boss will see only one race outcome, not the full distribution. Your task is to recommend a policy and explain why it is the right decision even if the single observed race makes it look wrong.

You have the same model Maya's team uses, in the accompanying workbook.

1. Open the simulator. Run a single race a few times and watch how differently the two policies can finish. Then run 1,000 paired races at once.
2. Compare the two policies on their full distribution of outcomes, not just their averages: the fast outcome, the median, the slow outcome, and how often each policy actually wins.
3. Decide which policy you would run, and state the conditions under which you would change your mind.
4. Be ready to defend your decision to a boss who has just watched it lose a race.